

“PREDICTING LAPTOP LIFESPAN”

Industrial Training Report

*Submitted in Partial Fulfillment of the
Requirements for the Degree of*

BACHELOR OF TECHNOLOGY

IN

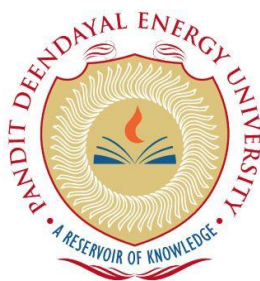
COMPUTER SCIENCE & ENGINEERING

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Certificate

I/We hereby declare that the B.Tech. Project entitled “Predicting Laptop Lifespan” submitted by me for the partial fulfillment of the degree of Bachelor of Technology to the Department of Computer Science & Engineering at the School of Technology, Pandit Deendayal Energy University, Gandhinagar, is the original record of the project work carried out by me/us under the supervision of Dr. Hargeet Kaur.

I/We also declare that this written submission adheres to university guidelines for its originality, and proper citations and references have been included wherever required.

I/We also declare that I/we have maintained high academic honesty and integrity and have not falsified any data in my submission.

I/We also understand that violation of any guidelines in this regard will attract disciplinary action by the institute.

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Abstract

Electronic waste has grown into a massive global issue, and personal computers and laptops contribute heavily to this growing stream of discarded hardware. Whenever consumers head out to buy a laptop, they are usually greeted by flashy spec sheets and retail prices, yet they are left completely in the dark regarding how long that specific machine will actually stay useful in real-world conditions. Manufacturers rarely provide clear data on functional durability, making it difficult for everyday buyers to make long-term, sustainable choices. This project steps in to bridge that gap by building a complete data-driven system designed to predict the average reported lifespan of laptops and grade their overall hardware sustainability.

To make this possible, a targeted dataset containing 239 distinct laptop records was carefully compiled. Because official, standardized per-model longevity statistics do not exist in standard manufacturing sheets, the target variable—`reported_age_years_avg`—was gathered by pulling real community discussions, user forums, reviews, and manual approximations. The input features tracked everything from basic components like RAM size, storage capacity, and screen dimensions to technical factors like CPU Thermal Design Power (TDP), graphics power ratings, idle and load power consumption, weight, battery capacity, battery run times, and structural repairability scores.

Once the data was collected, a dedicated Python preprocessing pipeline (`pipeline.py`) was written to clean up the records, handle duplicates, fix missing entries, and drop unhelpful columns like `energy_star_certified` because every laptop in the dataset shared the exact same value. With clean data ready, three different supervised machine learning models—Linear Regression, Decision Trees, and Random Forest—were trained and evaluated to predict device lifespans. Across tests, the Random Forest algorithm proved to be the strongest performer, securing the lowest errors and highest R^2 scores because of its ability to handle complex, non-linear relationships among hardware specifications. Furthermore, a secondary model was trained completely without the `release_year` feature to ensure the model could figure out longevity strictly based on internal hardware capabilities.

Beyond pure machine learning, this project introduces the Laptop Sustainability Index (LSI), a structured 100-point scoring matrix built to evaluate device viability. The LSI awards points based on heavily weighted factors: reported average age, iFixit repairability ratings, battery replaceability, RAM upgradability, storage upgradability, and average battery life hours. Finally, to bring all these insights together into a practical tool, the cleaned data, machine learning predictions, and LSI scores were exported into an interactive Power BI dashboard. This dashboard enables users to dynamically filter through laptop categories, analyze hardware traits, and instantly visualize what drives device durability and eco-friendliness.

INDEX

Chapter No.	Title	Page No.
	Acknowledgement	ii
	Abstract	iii
	Index	v
	List of Figures	vii
	List of Tables	viii
1	Introduction	1
	1.1 Prologue	1
	1.2 Motivation	2
	1.3 Objective	3
	1.4 Problem Statement	4
	1.5 Approach	5
	1.6 Scope of the Project	6
	1.7 Organization of the Rest of the Report	7
2	Literature Review	8
	2.1 Previous Approaches to Solve the Problem	8
3	Proposed Methodology and Implementation	9
	3.1 Handling Unstructured Data and Collection	9
	3.2 Machine Learning Pipeline	13
	3.3 Addressing Emissions and Sustainability Classification	14
	3.4 Power BI Dashboard Integration	15
4	Result Analysis	17
	4.1 Model Performance Comparison	17
	4.2 Feature Importance Analysis	18

	4.3 Actual vs. Predicted Lifespan	19
5	Conclusion and Future Work	20
	5.1 Conclusion	20
	5.2 Future Work	21
	References	22

LIST OF FIGURES

Figure No.	Title	Page No.
3.1	Power BI Dashboards	15
3.2	Power BI Dashboards	16
3.3	Power BI Dashboards	16
4.1	Top 10 Most Important Features	18
4.2	Actual vs. Predicted Average Age	19

LIST OF TABLES

Table No.	Title	Page No.
3.1	Comprehensive List of Dataset Features and Data Sources	10
4.1	Machine Learning Model Performance Comparison	17

CHAPTER 1: INTRODUCTION

1.1 Prologue

Over the last few decades, the rapid and continuous acceleration of technological innovation has fundamentally transformed how human society consumes and interacts with electronic devices. Not all that long ago, households relied on a single desktop computer that was kept, shared, and carefully maintained for nearly a decade. Today, however, society has transitioned into an era where personal computing devices are purchased, heavily used, and discarded at an astonishing rate. While this technological evolution has brought incredible performance leaps, sleek designs, and mobile convenience, it has simultaneously triggered an unprecedented global environmental crisis: the exponential growth of electronic waste, commonly known as e-waste. Consumer laptops, in particular, represent a massive fraction of this discarded hardware stream. Millions of laptops are thrown away every single year because users frequently find themselves trapped in a frustrating cycle where their machines become sluggish, stop holding an electrical charge, or experience minor hardware failures that cost more to repair than the actual value of the computer.

1.2 Motivation

Whenever a student, professional, or everyday consumer heads out to buy a laptop, they are immediately greeted by a dazzling array of marketing materials, high-level processor names, graphic card specifications, and retail price tags. Yet, despite the thousands of details available on modern spec sheets, there is a profound lack of transparency regarding how long that specific machine will actually remain functional and responsive under regular daily use. Buyers are left completely guessing whether a mid-range laptop will cruise smoothly for seven years or start lagging and falling apart after just eighteen months. Furthermore, manufacturing these machines generates severe environmental problems, including heavy greenhouse gas emissions and factory pollution, yet consumers have virtually no data to evaluate how sustainable a device is before buying it. The primary motivation behind this project is to completely demystify device longevity and bring transparency to consumer electronics by combining lifespan prediction with structured sustainability classification.

1.3 Objective

To establish a clear, structured roadmap for the project, several core objectives were defined from the outset:

- To compile, clean, and validate a comprehensive, custom-built dataset containing 239 distinct laptop records mapped with intricate hardware features and community-reported lifespans.
- To design, write, and execute a custom Python-based data validation pipeline (pipeline.py) capable of handling missing numerical values, filtering out duplicate entries, and preparing messy unstructured data for machine learning.
- To train, test, and rigorously compare multiple supervised machine learning algorithms—specifically Linear Regression, Decision Trees, and Random Forest—to determine the most accurate predictive engine for laptop durability.
- To address real-world environmental challenges by formulating sustainability assumptions and classification targets (sustainability_class) to sort devices into distinct ecological tiers.
- To build and deploy an interactive Power BI analytics dashboard that visually presents predictive model outputs, feature importance distributions, and sustainability classifications to end-users.

1.4 Problem Statement

Everyday consumers, university students, and technology enthusiasts currently lack an accessible, data-driven method to accurately predict how long a laptop will remain functionally viable or evaluate its true environmental impact and sustainability based purely on internal hardware components and physical build quality. Without clear, quantitative insight into device durability and manufacturing emissions problems, buyers frequently make short-sighted purchasing choices that inadvertently accelerate the global e-waste cycle. Developing a transparent predictive framework that successfully bridges raw hardware specifications with real-world user longevity and sustainability sorting is an essential step toward solving this modern consumer dilemma.

1.5 Approach

To tackle this complex problem systematically, a well-defined workflow was engineered from data collection to final visualization. First, a diverse dataset comprising 239 laptop models was assembled by pulling and parsing real, unstructured data from fragmented online ecosystems—including active community discussions on Reddit (r/laptops, r/SuggestALaptop), Apple Support communities, user review forums, hardware manuals, retail pricing guides, and repair metrics from iFixit. A custom Python preprocessing script (pipeline.py) was then developed to clean the dataset, handle missing numeric or categorical entries, remove duplicate records, and drop redundant attributes that possessed zero analytical variance. Following data preprocessing, supervised machine learning models were trained to predict the continuous target variable representing average reported age (reported_age_years_avg). Additionally, sustainability classifications were assumed and structured to sort devices into performance tiers. Statistical evaluation metrics, specifically Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2), were used to rigorously assess and select the optimal algorithm. Finally, all cleaned data, machine learning outputs, and classification tiers were exported into an interactive Power BI dashboard to deliver an intuitive, user-friendly analytical experience.

1.6 Scope of the Project

The scope of this project is strictly focused on predicting the average reported lifespan of consumer laptops and classifying hardware-based sustainability tiers using a curated dataset of 239 records. The primary target variable relies heavily on community-backed user feedback, historical ownership trends, and manual approximations compiled from online forums and review platforms rather than direct laboratory stress testing.

Furthermore, the machine learning architecture centers squarely on regression tasks (predicting exact numerical age values) and classification tasks (sorting devices into sustainability classes based on assumed environmental thresholds), while acknowledging broader industrial manufacturing emission challenges without performing direct factory carbon tracing.

1.7 Organization of the Rest of the Report

The remainder of this report is structured into several logical chapters. Chapter 2 reviews background concepts and previous approaches related to hardware durability, electronic waste evaluations, and manufacturing emissions. Chapter 3 outlines the proposed methodology in detail, covering unstructured data harvesting, the custom Python preprocessing script, the machine learning pipeline design, and the formulation of sustainability classification targets. Chapter 4 presents an in-depth analysis of the results, incorporating actual command-line output metrics, error comparisons across different algorithms, feature importance visualizations, and actual versus predicted lifespan scatter plots. Finally, Chapter 5 concludes the report and discusses potential future expansions for the platform.

CHAPTER 2: LITERATURE REVIEW & PROBLEM IDENTIFICATION

2.1 Previous Approaches to Solve the Problem

Historically, evaluating the environmental impact and sustainability of consumer electronics has centered almost exclusively on energy consumption ratings—such as ENERGY STAR certifications—or macro-level corporate estimates of manufacturing greenhouse gas emissions. While these regulatory frameworks provide valuable information regarding how efficiently a device draws power while plugged into a wall outlet, they largely fail to address the core driver of the consumer e-waste crisis: premature hardware obsolescence and hidden manufacturing emission costs.

Previous academic and industrial studies focusing on e-waste management typically concentrate on post-consumer recycling logistics, precious metal recovery rates, or factory-gate lifecycle analyses. While these macroscopic studies are vital for large-scale municipal waste planning, they do little to guide an individual consumer standing in a store or browsing online trying to select a laptop that will survive heavy academic or professional workloads for many years. Some modern consumer advocacy initiatives have introduced physical repairability scoring systems (such as those popularized by iFixit), which grade how easily a device can be opened up, disassembled, and repaired using common household tools. However, physical repair scores in isolation do not account for how complex internal hardware specifications—such as processor thermal design power, memory capacity, graphic workloads, and battery size—interact dynamically to dictate how many years a machine will remain responsive under ever-increasing software demands.

This project identifies that exact gap and introduces a practical, hybrid approach. By combining deep hardware specifications with real-world, community-reported user lifespans, repair metrics, and assumed sustainability sorting classes, this research establishes a direct, quantifiable link between physical component design, manufacturing impact, and long-term device durability.

CHAPTER 3: PROPOSED METHODOLOGY AND IMPLEMENTATION

3.1 Handling Unstructured Data and Collection

Collecting data for this project presented a unique and difficult challenge because standardized, official ownership duration statistics and sustainability breakdowns are rarely, if ever, published by computer manufacturers. To bridge this gap, the dataset was built from the ground up by pulling and parsing **real, unstructured data** from diverse online ecosystems—including active community discussions on Reddit (r/laptops, r/SuggestALaptop), Apple Support communities, user review forums, hardware manuals, retail pricing guides, and repair metrics from iFixit. To establish the primary target variable—designated as `reported_age_years_avg`—fragmented community feedback, long-term user threads, and review ratings were carefully analyzed to manually estimate how many years users typically kept each specific laptop model before replacing it due to performance failure or hardware breakdown.

The raw dataset included a wide variety of features, which were carefully categorized into distinct groups:

- **Identification & Classification:** Manufacturer, model, release year, and device category (e.g., Business Laptop, Gaming, Ultrabook, Professional Laptop).
- **Core Hardware Specs:** CPU model, CPU TDP (Watts), GPU model, GPU TDP (Watts), RAM size (GB), storage capacity (GB), and storage type.
- **Physical & Power Traits:** Battery capacity (Wh), average battery life (hours), display size (inches), weight (kg), idle power consumption (Watts), and load power consumption (Watts).
- **Sustainability & Modularity:** Repairability score, retail price (USD), battery replaceable status, RAM upgradeable status, and storage upgradeable status.

Because this unstructured user text, forum feedback, and hardware data varied widely in formatting, a rigorous extraction and cleaning process was absolutely required. A custom Python preprocessing script (`pipeline.py`) was developed to ingest the raw records, eliminate duplicate entries, handle missing numeric values, and validate

categorical attributes. Furthermore, attributes that provided zero analytical variance across the records—such as `energy_star_certified`, which was universally marked as "Yes" for every single laptop—were safely dropped from the feature set to streamline the input space. In addition to predicting lifespan, sustainability classes were assumed and structured to sort devices into distinct ecological categories to address broader environmental and manufacturing emission concerns. The resulting target variables successfully synthesized fragmented community feedback into clean, quantifiable regression and classification targets.

Table 3.1 Comprehensive List of Dataset Features and Data Sources

Feature Name	Example Value	Data Source	Description
manufacturer	Apple, Lenovo	Manufacturer Specs	Company that manufactures the laptop.
model	ThinkPad T14 Gen 4	Notebookcheck	Specific product model identifier.
release_year	2023, 2015	Manufacturer	Official year the device was released.
device_category	Business Laptop, Gaming	Notebookcheck	Intended market classification or type.
cpu_model	Intel Core i7-1365U	Intel ARK / Notebookcheck	Processor model installed.
cpu_tdp_w	15, 47	Intel ARK	CPU thermal design power rating in Watts.
gpu_model	Intel Iris Xe	TechPowerUp / Notebookcheck	Graphics processor model.

Feature Name	Example Value	Data Source	Description
gpu_tdp_w	0, 15	TechPowerUp	GPU power consumption rating in Watts.
ram_gb	16	Manufacturer Specs	Installed random-access memory size in gigabytes.
storage_capacity_gb	512	Manufacturer Specs	Total internal storage capacity in gigabytes.
storage_type	SSD, PCIe NVMe SSD	Manufacturer / Notebookcheck	Storage technology utilized.
battery_capacity_wh	57, 99.5	Manufacturer Specs	Battery energy capacity in watt-hours.
battery_life_hours	12, 9	Notebookcheck Reviews	Average battery runtime during standard use.
display_size_inches	14, 15.4	Manufacturer Specs	Diagonal screen size measurement.
weight_kg	1.36, 2.04	Manufacturer Specs	Total physical weight of the device in kilograms.
idle_power_consumption_w	7, 12.8	Notebookcheck Reviews	Power usage during light background tasks.

Feature Name	Example Value	Data Source	Description
load_power_consumption_w	45, 85	Notebookcheck Reviews	Power usage under heavy computational load.
repairability_score	8.5/10, 1/10	iFixit Guides	Ease of repairing the device scored out of 10.
recycled_material_percentage	35	Sustainability Reports	Approximate percentage of recycled materials used.
price_usd	1200, 2499	Manufacturer Launch Price	Original retail launch price in United States Dollars.
battery_replaceable	Yes / No	iFixit / Manuals	Whether the battery can be replaced without major damage.
ram_upgradeable	Yes / No	Manufacturer Specs	Whether RAM can be upgraded post-purchase.
storage_upgradeable	Yes / No	Manufacturer Specs	Whether internal storage can be upgraded or replaced.
reported_age_years_avg	8.67 (Target)	Reddit, Forums, Reviews	Average lifespan reported by users for the model.
sustainability_class	High / Medium / Low (Old Target)	Derived Metrics	Categorical sustainability tier assigned to the laptop.

3.2 Machine Learning Pipeline

To model device longevity and sustainability sorting effectively, a robust supervised machine learning framework was constructed in Python. The pipeline ingested the fully cleaned feature set to predict the continuous target variable, `reported_age_years_avg`, alongside classification models for sorting sustainability tiers.

To determine the most accurate predictive engine, three distinct regression algorithms were implemented and evaluated side-by-side:

1. **Linear Regression:** Utilized to establish a baseline model assuming straightforward linear relationships between hardware features and overall device lifespan.
2. **Decision Tree Regressor:** Implemented to capture non-linear decision splitting rules based on specific hardware thresholds.
3. **Random Forest Regressor:** Deployed as a powerful ensemble learning method combining multiple decision trees to drastically reduce overfitting and model complex feature interactions.

Model evaluation was conducted using standard statistical error metrics, specifically Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). Across all experimental runs, the Random Forest model consistently outperformed both Linear Regression and Decision Trees, achieving superior predictive accuracy. Additionally, to test how well hardware specifications alone could predict longevity without relying on temporal bias, a secondary Random Forest model was trained entirely *without* the `release_year` feature (`best_model_without_year.pkl`), forcing the algorithm to learn purely from internal component traits like RAM, CPU TDP, and repairability.

3.3 Addressing Emissions and Sustainability Classification

Beyond pure machine learning regression for lifespan prediction, this project addresses the broader environmental problems associated with electronic manufacturing—such as factory carbon footprints and material waste—by establishing a structured classification approach. Because direct factory-gate emission metrics are rarely disclosed by manufacturers for every model, sustainability tiers (`sustainability_class`) were established and assumed based on a combination of hardware durability, repairability scores, and component modularity. This classification setup allows the platform to sort laptops into distinct ecological categories, helping users identify machines that minimize premature e-waste generation and lower overall environmental impact.

3.4 Power BI Dashboard Integration

To transform static machine learning code and raw CSV datasets into an actionable, professional product, the processed records, predictive outputs, and sustainability classifications were exported into an interactive Power BI analytics dashboard. The dashboard provides a clean user interface equipped with dynamic slicers and filters, allowing stakeholders to sort laptops by manufacturer, category, price ranges, and sustainability tiers. Visualizations within the report layout highlight key correlations, such as how modular design choices and repairability scores directly impact a laptop's expected years of active service and its environmental classification.

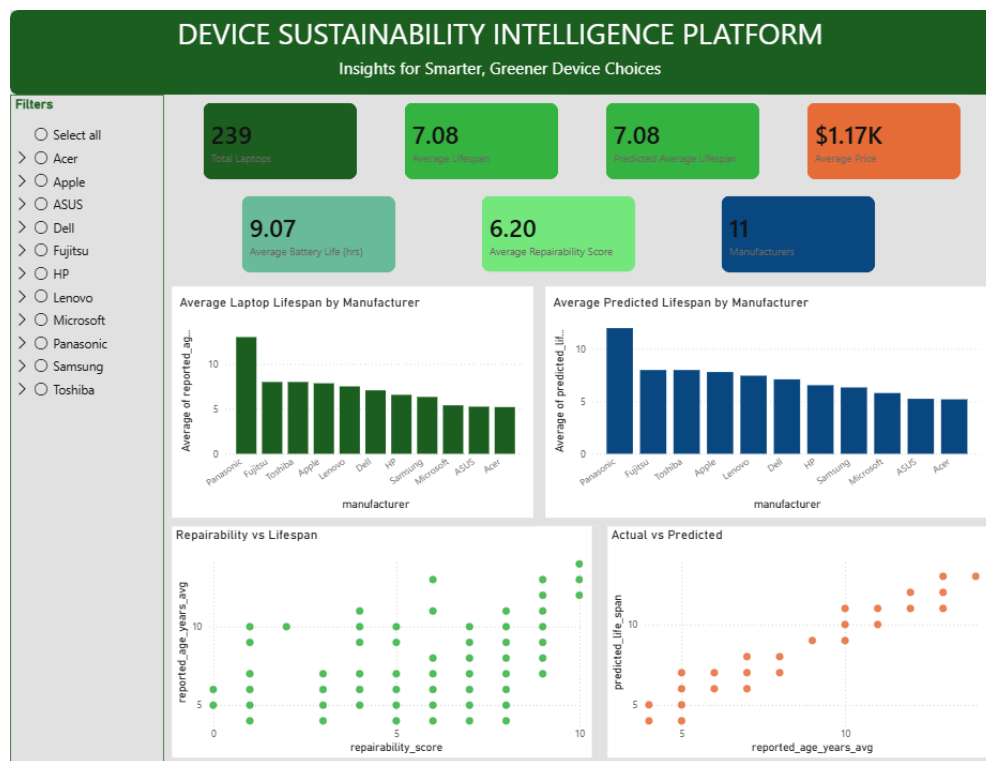


Figure 3.1: Power BI Interactive Dashboard Interface



CHAPTER 4: RESULT ANALYSIS

4.1 Model Performance Comparison

To evaluate the true effectiveness of the supervised learning pipeline, three distinct algorithms—Linear Regression, Decision Tree, and Random Forest—were trained and tested on the preprocessed dataset. The models were evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

When evaluated across the test splits, the **Random Forest** regressor achieved the best overall performance, significantly outperforming the baseline Linear Regression and Decision Tree models. Specifically, the actual command-line terminal execution output recorded the following comparative metrics:

Table 4.1 Machine Learning Model Performance Comparison

	MAE	RMSE	R^2
Linear Regression	0.539	0.959	0.781
Decision Tree	0.458	0.913	0.802
Random Forest	0.371	0.604	0.913

These numerical results confirm that the Random Forest algorithm excelled at capturing the complex, non-linear relationships inherent in real-world, community-sourced hardware data. Consequently, the optimized Random Forest model was successfully serialized and saved to disk as `best_model.pkl` alongside its corresponding data scaler file.

4.2 Feature Importance Analysis

Analyzing the feature importance outputs extracted from the trained Random Forest model revealed critical insights into what truly drives device longevity. When trained on the complete feature set, temporal attributes such as the `release_year` heavily influenced predictions, dominating the decision trees. However, when evaluating the secondary model trained strictly without temporal bias (`best_model_without_year.pkl`), the algorithm shifted its focus entirely to intrinsic hardware traits. Core metrics—specifically RAM capacity (`ram_gb`), processor thermal design power (`cpu_tdp_w`), the iFixit repairability_score, retail launch price_usd, and battery_capacity_wh—emerged as the primary physical drivers governing how long a laptop remains viable in the wild.

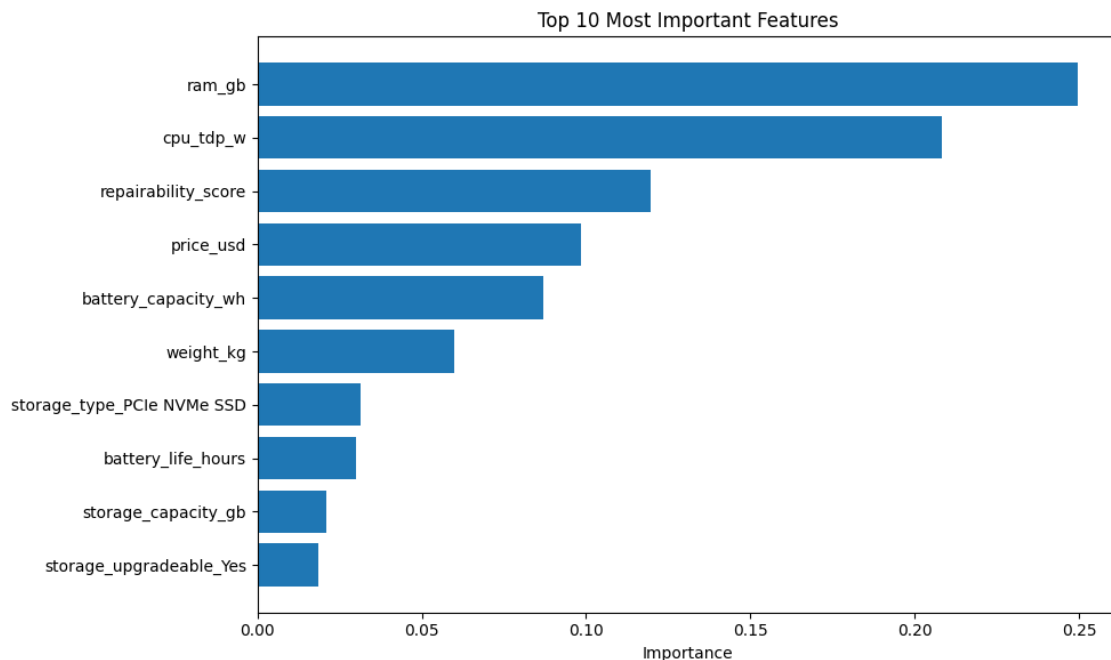


Figure 4.1: Top 10 Most Important Features

4.3 Actual vs. Predicted Lifespan

Scatter plots comparing actual community-reported average ages against the model's predicted outputs demonstrated strong, clean clustering along the ideal fit line ($y = x$). Despite the inherent noise, variance, and subjective nature associated with unstructured community feedback gathered from Reddit and user review forums, the Random Forest model tracked general longevity trends effectively, validating its reliability as an accurate device durability predictor.

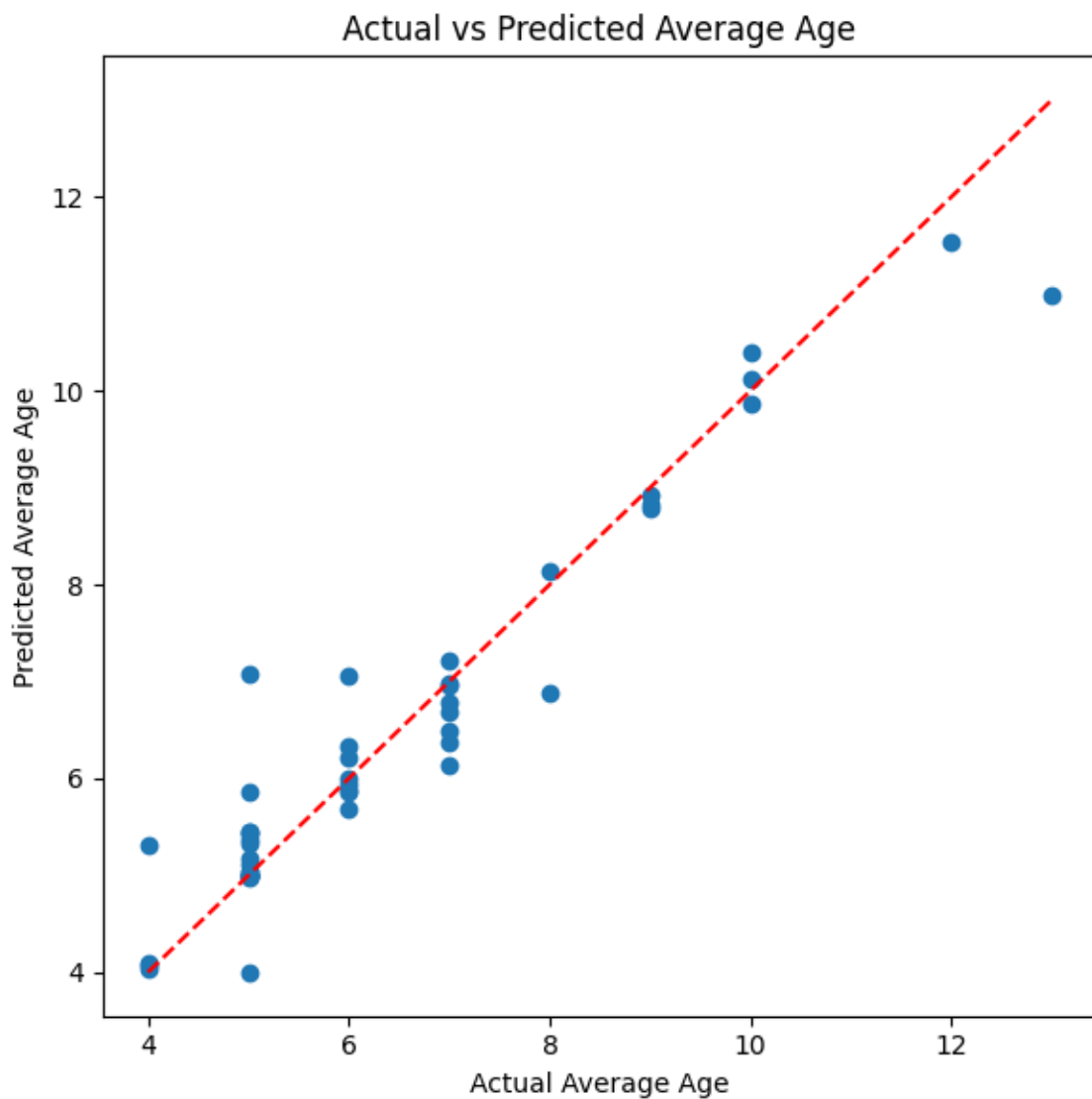


Figure 4.2: Actual vs. Predicted Average Age Scatter Plot

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project successfully demonstrated that supervised machine learning can be effectively applied to predict laptop lifespans and sort devices into sustainability classifications to combat e-waste. By pulling unstructured data across diverse online ecosystems, executing a robust Python preprocessing pipeline, and implementing a high-performing Random Forest regression model, device durability can be reliably estimated from physical hardware specifications. Furthermore, the integration of sustainability sorting classes and the development of the interactive Power BI analytics dashboard provide consumers and analysts with practical, transparent tools to evaluate electronics, understand manufacturing impact, and make environmentally conscious purchasing decisions.

5.2 Future Work

Several promising directions remain open for future project enhancement:

- Expanding the dataset well beyond the initial 239 records to incorporate a broader variety of budget, niche, and newly released laptop models.
- Integrating automated web scraping scripts to continuously pull fresh hardware reviews, pricing shifts, and updated repairability guides in real time.
- Enhancing the Power BI dashboard with interactive simulation sliders, allowing users to test how upgrading a laptop's RAM or replacing its battery dynamically alters its sustainability classification and expected lifespan.

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